

BCSE0131: DIGITAL IMAGE PROCESSING LAB

Objective: The objective of this Lab is to develop hands-on experience to write programs using MATLAB/Python language for digital manipulation of images in both spatial and frequency domains.

Credits: 01

L-T-P-J: 0-0-2-0

Module No.	Content	Lab Hours
I/II	- Basic commands to familiarize with MATLAB & perform the various Matrix operations. - Understanding image basic "image resize, image type conversion, extraction of color band, creating a synthetic image, pseudocolor image" - Perform various arithmetic operation (image addition, subtraction & complement) & logical operation (NOT, OR and XOR) on images - Perform various Image Enhancement operations: Image Negation function, Logarithmic Transformation, Power Law Transformation, Histogram Equalization, contrast stretching, plot histogram without using imhist function - Perform smoothing using linear (average filter) and order statistics filters (min, max & median) of varying sizes - Sharpen an image using Laplacian filter. - Perform various Fast Fourier transform (FFT) and frequency domain filtering on images using MATLAB. - Perform various Image Enhancement operation in frequency domain - Perform various Morphological operation dilation, erosion, internal & external boundary Extraction, Thinning, thickening of image & Perform Dilation, erosion, boundary Extraction without using direct function - Perform various thresholding segmentation (Simple, Multiple, and Adaptive thresholding) - Perform the various Edge Detection Operators (Ordinary, Roberts, Prewit, Sobel and Canny Operator) - Minor Application Assignment.	12*2=24

Reference Books:

- R.C. Gonzalez and R.E. Woods, "Digital Image Processing Using MATLAB", PHI, 2nd Edition, 2010.
- Hands-On Image Processing with Python by Sandipan Dey, November 2018, Packt
- <https://www.pyimagesearch.com/>

Outcome: After studying the subject, the students will be able to:

- CO1: Implement digital image processing operations for image manipulation and Enhancement.
- CO2: Test the source code for Morphological and Segmentation algorithms.
- CO3: Apply image processing algorithms in real-life problems as Minor Application Assignment.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1, PO2, PO6, PO9/PSO1, PSO2
CO2	PO5, PO7, PO8, PO10/PSO1, PSO4
CO3	PO1, PO3, PO11, PO12/PSO3, PSO4